Harikrishna Trivedi

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EDUCATION

New Jersey Institute of Technology (NJIT) Newark, NJ
Master of Science, Computer Science - Expected May 2028 *Ongoing*

- Focus Areas: Machine Learning, Natural Language Processing, Agentic AI Systems, Computer Vision

Pandit Deendayal Energy University (PDEU) Gandhinagar, India
B.Tech, Electronics & Communication Engineering - CGPA: 7.4 / 10 *May 2023 – May 2026*

- Relevant: Machine Learning, Data Structures & Algorithms, Python Programming, Digital Signal Processing, MATLAB
- Final year project: Fabricated a 3-Band MIMO Antenna with Multifrequency

TECHNICAL SKILLS

AI / ML: PyTorch, TensorFlow, Scikit-learn, Keras, OpenCV, MediaPipe, NumPy, Pandas, Matplotlib
LLM & Agents: GPT-4o / 4o-mini API, Gemini, LangChain, n8n Workflow Automation, Prompt Engineering
Languages: Python, JavaScript, SQL, MATLAB, C#, C++
Cloud & MLOps: Google Cloud Platform (GCP), Vertex AI, AWS, Docker, CI/CD, REST APIs, Git
Databases: PostgreSQL, MongoDB, MySQL, Microsoft SQL Server

PROJECTS

MedIntel AI - LLM-Powered Medical Report Explainer 2025
FastAPI · GPT-4o-mini · Next.js · React · TailwindCSS · Tesseract OCR · PostgreSQL

- Built end-to-end AI pipeline: Tesseract OCR extracts clinical text from uploaded PDFs; GPT-4o-mini converts medical jargon to structured plain-language patient summaries via engineered prompts with JSON schema enforcement
- Designed async FastAPI backend with streaming responses; React frontend with multi-step loading states and collapsible result cards handles 100+ concurrent uploads at sub-2s first-token latency
- Authored structured prompt chain separating diagnosis parsing, risk explanation, and recommended-action generation - achieving consistent output quality across diverse report formats

Healthcare Disease Prediction System Jan 2024 – May 2024
Python · TensorFlow · PyTorch · Scikit-learn · Flask · Pandas · NumPy

- Engineered ensemble ML pipeline (Random Forest + Gradient Boosting) for early detection of diabetes, heart disease, and hypertension; achieved 87% accuracy on 10,000+ patient records
- Applied feature engineering, normalization, and GridSearchCV hyperparameter tuning to improve model F1 score by 15%; authored research paper on findings under faculty supervision
- Deployed via Flask REST API with interactive web dashboard enabling real-time disease risk assessment for clinical staff

Sign Language to Speech Conversion System Aug 2023 – Dec 2023
Python · PyTorch · CNN · OpenCV · MediaPipe · Google Text-to-Speech API

- Trained CNN on 50,000+ labeled ASL gesture images with data augmentation; achieved 92% classification accuracy across 26 gestures at 30 FPS - reduced overfitting by 23% via dropout and augmentation
- Implemented MediaPipe hand-tracking for real-time gesture segmentation; integrated Google TTS API for seamless ASL-to-audio translation serving hearing-impaired users

PROFESSIONAL EXPERIENCE

Vedlogic Solutions Pvt Ltd India
AI Intern *Jan 2026 – Jun 2026*

- Engineered an end-to-end HR onboarding automation system using n8n, Nginx, and Weam AI; workflow was adopted into the organization's core operations - replacing fully manual employee onboarding and HR processing pipelines
- Designed multi-trigger n8n pipelines integrating Weam AI agents for intelligent document classification and automated task routing across onboarding stages; configured Nginx as reverse proxy for secure, self-hosted internal deployment

- Delivered a production-ready automation stack that required zero post-handoff rework - intern-level contribution absorbed directly into the org's active tooling

Vohra Wound Physicians India

Junior Full-Stack Developer *Jan 2023 – Jun 2023*

- Shipped 15+ production features and resolved 50+ critical bugs in a HIPAA-sensitive healthcare management platform using .NET Framework, C#, and jQuery - improved application stability by 35%
- Integrated RESTful APIs with Microsoft SQL Server for patient data management; conducted unit and integration testing that surfaced 80+ pre-production issues and achieved zero critical bugs across all releases